
Linear Statistical Models

B. Statistical Data Science 2nd Year Indian Statistical Institute

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Revision 4 – Linear Algebra

Exercise 1. We saw in the class that for any subspace $\mathcal{S} \subseteq \mathbb{R}^n$, there exists a unique projection matrix onto $\mathcal{S}$ that is symmetric. We call this the **orthogonal projection** onto $\mathcal{S}$ and denote it by $P_{\mathcal{S}}$. We also saw that if $\{\mathbf{u}_1, \dots, \mathbf{u}_r\}$ is an orthonormal basis of $\mathcal{S}$, then $P_{\mathcal{S}} = \sum_{i=1}^r \mathbf{u}_i \mathbf{u}_i^{\top}$ is the orthogonal projection onto $\mathcal{S}$. Since $P_{\mathcal{S}}$ is unique, it follows that $\sum_{i=1}^r \mathbf{u}_i \mathbf{u}_i^{\top}$ DOES NOT depend on the particular choice of orthonormal basis. We will verify this differently here.

- (a) Show that $\sum_{i=1}^r \mathbf{u}_i \mathbf{u}_i^{\top} = \mathbf{U} \mathbf{U}^{\top}$, where $\mathbf{U} = (\mathbf{u}_1, \dots, \mathbf{u}_r)$. Observe that the columns of $\mathbf{U}$ are orthonormal, so that $\mathbf{U}$ is an orthogonal matrix.
- (b) Consider a different orthonormal basis $\{\mathbf{v}_1, \dots, \mathbf{v}_r\}$ of $\mathcal{S}$. Define $\mathbf{V} = (\mathbf{v}_1, \dots, \mathbf{v}_r)$. Show that there exists a square matrix $\mathbf{W}$ such that $\mathbf{U} = \mathbf{V} \mathbf{W}$.
- (c) Show that $\mathbf{W}$ in the previous step satisfies $\mathbf{W}^{\top} \mathbf{W} = \mathbf{I}_r$.

Hint: The columns of $\mathbf{U}$ are orthonormal. The same is true for $\mathbf{V}$.

- (d) Show that the previous part implies $\mathbf{W}$ is invertible and $\mathbf{W}^{-1} = \mathbf{W}^{\top}$.
- (e) Show that $\mathbf{U} \mathbf{U}^{\top} = \mathbf{V} \mathbf{V}^{\top}$.

Exercise 2. Here we check some facts about orthogonal projection matrices.

- (a) Show that the only non-singular orthogonal projection matrix is the identity matrix $\mathbf{I}$. Hence, any orthogonal projection that is not $\mathbf{I}$ must be singular. In particular, orthogonal projections onto proper subspaces are **rank deficit** (not of full rank).
- (b) Let $\mathcal{S}_1, \mathcal{S}_2$ be two subspaces of $\mathbb{R}^n$. Recall that $\mathcal{S}_1$ and $\mathcal{S}_2$ are said to be orthogonal, denoted $\mathcal{S}_1 \perp \mathcal{S}_2$, if $\mathbf{s}_1 \perp \mathbf{s}_2$ for every $\mathbf{s}_1 \in \mathcal{S}_1, \mathbf{s}_2 \in \mathcal{S}_2$. Show that $\mathcal{S}_1 \perp \mathcal{S}_2$ if and only if $P_{\mathcal{S}_1} P_{\mathcal{S}_2} = \mathbf{O}$.
- (c) Show that $\mathcal{S}_1 \subseteq \mathcal{S}_2$ if and only if $P_{\mathcal{S}_1} P_{\mathcal{S}_2} = P_{\mathcal{S}_2} P_{\mathcal{S}_1} = P_{\mathcal{S}_1}$.

Note: For two subspaces $\mathcal{S}_1$ and $\mathcal{S}_2$, $P_{\mathcal{S}_2} P_{\mathcal{S}_1} \mathbf{x}$ is the result of first projecting $\mathbf{x}$ onto $\mathcal{S}_1$ and then projecting the result onto $\mathcal{S}_2$. In general, $P_{\mathcal{S}_2} P_{\mathcal{S}_1}$ and $P_{\mathcal{S}_1} P_{\mathcal{S}_2}$ need not be the same.

- (d) Show that if $\mathcal{S}_1$ and $\mathcal{S}_2$ are orthogonal subspaces of $\mathbb{R}^n$, then $P_{\mathcal{S}_1 \oplus \mathcal{S}_2} = P_{\mathcal{S}_1} + P_{\mathcal{S}_2}$.

Note: Is orthogonality of $\mathcal{S}_1$ and $\mathcal{S}_2$ necessary for this result?

- (e) Show that if $\mathcal{S}_1 \subseteq \mathcal{S}_2$, then $P_{\mathcal{S}_2} - P_{\mathcal{S}_1}$ is also an orthogonal projection. So, the difference of two orthogonal projections can be an orthogonal projection. Where does it project onto?
- (f) Let $\mathbf{X}^{n \times p} = (\mathbf{X}_1^{n \times q}, \mathbf{X}_2^{n \times r})$ be of full column rank. Show that $\mathbf{X}(\mathbf{X}^{\top} \mathbf{X})^{-1} \mathbf{X}^{\top} - \mathbf{X}_1(\mathbf{X}_1^{\top} \mathbf{X}_1)^{-1} \mathbf{X}_1^{\top}$ is an orthogonal projection. Where does it project onto?

Note: Observe that $\mathbf{X}$ having full column rank implies both $\mathbf{X}^{\top} \mathbf{X}$ and $\mathbf{X}_1^{\top} \mathbf{X}_1$ are invertible.

Exercise 3. When dealing with column spaces of a matrix A , i.e., when $\mathcal{S} = \mathcal{C}(A)$, we will write P_A (instead of $P_{\mathcal{C}(A)}$) to denote the orthogonal projection onto $\mathcal{C}(A)$. Let A and B be two matrices with the same number of rows. We write $[A, B]$ to denote the matrix obtained by stacking A and B side-by-side.

- (a) Show that $\mathcal{C}(A) \perp \mathcal{C}((I - P_A)B)$.
- (b) Show that $\mathcal{C}(A) + \mathcal{C}((I - P_A)B) = \mathcal{C}([A, B])$. Conclude that $\mathcal{C}([A, B]) = \mathcal{C}(A) \oplus \mathcal{C}((I - P_A)B)$.
- (c) Show that $P_{[A, B]} = P_A + P_{(I - P_A)B}$.

Note: The last result can be seen as orthogonalization for subspaces. Observe the similarity with the Gram-Schmidt procedure.